Principal G-bundles over X are in Natural Bijective Correspondence with $[X, BG]$

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Abstract

We present the classical result that isomorphism classes of principal G-bundles over a CW complex X are in natural bijective correspondence with homotopy classes of maps $[X, K(G, 1)]$, where G is a discrete group and $K(G, 1)$ denotes the Eilenberg–MacLane space.

1 Principal G- bundles

Definition 1.1 (Principal G-bundle)**.** *Let G be a topological group and P be a right G-space. A continuous map*

$$\pi : P \to B$$

*is called a **principal G-bundle** if the action is fiber-preserving and satisfies local triviality:*

1. ***Fiber-preserving:*** *For every $g \in G$ and $p \in P$,*

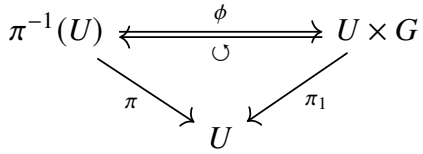
$$\pi(p \cdot g) = \pi(p).$$

2. ***Local triviality:*** *For every $b \in B$, there exists an open neighborhood $U \subseteq B$ of b and a homeomorphism*

$$\phi : \pi^{-1}(U) \to U \times G$$

such that:

- *$\pi_1 \circ \phi = \pi|_{\pi^{-1}(U)}$, where $\pi_1 : U \times G \to U$ is projection onto the first factor.*



- *ϕ is G-equivariant, i.e., for all $p \in \pi^{-1}(U)$ and $g \in G$,*

$$\phi(p \cdot g) = \phi(p) \cdot g,$$

where the action on $U \times G$ is given by

$$(u, h) \cdot g = (u, hg).$$

1.1 Properties of Principal G-bundles

Action on Fibers

For each $b \in B$, consider the fiber

$$P_b = \pi^{-1}(\{b\}).$$

1. **The action of G on P_b is free and transitive.**

Proof. We prove freeness and transitivity separately.
Free: Suppose $p \in P_b$ and $g \in G$ such that $p \cdot g = p$. We want to show $g = e$ (the identity element).

Choose a local trivialization $\phi : \pi^{-1}(U) \to U \times G$ around $b \in U$. Let $\phi_b = \phi|_{P_b} : P_b \to \{b\} \times G$ be the restriction to the fiber. Suppose $\phi_b(p) = (b, h)$ for some $h \in G$.

By G-equivariance of ϕ, we have:

$$\phi_b(p \cdot g) = \phi_b(p) \cdot g = (b, h) \cdot g = (b, hg).$$

Since $p \cdot g = p$, we obtain:

$$\phi_b(p \cdot g) = \phi_b(p) \implies (b, hg) = (b, h).$$

Therefore, $hg = h$, and multiplying on the left by h^{-1} yields $g = e$. Thus, the action is **free**.
Transitive: Let $p, p' \in P_b$ be any two points in the same fiber. Again, using the local trivialization $\phi_b : P_b \to \{b\} \times G$, we have:

$$\phi_b(p) = (b, h), \quad \phi_b(p') = (b, h')$$

for some $h, h' \in G$. Let $g = h^{-1}h' \in G$. Then:

$$\phi_b(p) \cdot g = (b, h) \cdot g = (b, hg) = (b, hh^{-1}h') = (b, h') = \phi_b(p').$$

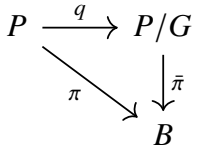
Since ϕ_b is a bijection and G-equivariant, we conclude that $p \cdot g = p'$. Thus, the action is **transitive**. □

2. **The orbit space P/G is homeomorphic to the base space B.**

Proof. The projection map $\pi : P \to B$ is constant on each G-orbit by the fiber-preserving property:

$$\pi(p \cdot g) = \pi(p) \quad \text{for all } p \in P, g \in G.$$

Therefore, π factors through the quotient map $q : P \to P/G$, yielding a continuous map $\bar{\pi} : P/G \to B$ defined by $\bar{\pi}([p]) = \pi(p)$.



We claim $\bar{\pi}$ is a homeomorphism.
Well-defined and continuous: Clear from the universal property of quotient spaces.
Injectivity: If $\bar{\pi}([p]) = \bar{\pi}([p'])$, then $\pi(p) = \pi(p') = b$. Thus $p, p' \in P_b$. By transitivity of the action on fibers, there exists $g \in G$ such that $p' = p \cdot g$. Hence $[p] = [p']$ in P/G.
Surjectivity: Since π is surjective, for any $b \in B$, there exists $p \in P$ with $\pi(p) = b$. Then $\bar{\pi}([p]) = b$.
Open map / Continuity of inverse: Local triviality of $P \to B$ implies that locally, $P|_U \cong U \times G$. Passing to the quotient by G, we get $(P|_U)/G \cong U \times (G/G) \cong U$. Thus $\bar{\pi}$ is locally a homeomorphism, hence a global homeomorphism. □

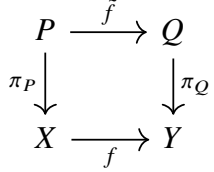
Summary: For a principal G-bundle $\pi : P \to B$:

- Each fiber P_b is a G-torsor (free and transitive G-space).
- The base space B is naturally isomorphic to the orbit space P/G.

Definition 1.2 (Morphism of principal G-bundles)**.** *A **morphism** between two principal G-bundles $\pi_P : P \to X$ and $\pi_Q : Q \to Y$ is a pair $(f, \tilde{f})$ consisting of:*

- *a continuous map $f : X \to Y$ (the base map),*
- *a continuous map $\tilde{f} : P \to Q$ (the bundle map),*

such that the following diagram commutes:



and $\tilde{f}$ is G-equivariant, i.e., $\tilde{f}(p \cdot g) = \tilde{f}(p) \cdot g$ for all $p \in P$, $g \in G$.

Proposition 1.3. *Every morphism of principal G-bundles over the same base space B with identity base map $\mathrm{id}_B : B \to B$ is an isomorphism.*

Proof. Let $\tilde{f} : P \to Q$ be a morphism of principal G-bundles over B with base map id_B. Suppose first that both bundles are trivial, i.e., $P = Q = B \times G$. Then the bundle map has the form:

$$\tilde{f}(b, g) = (b, \rho(b)g)$$

for some function $\rho : B \to G$. Since $\tilde{f}$ is continuous, ρ is continuous. The inverse map is then given by:

$$\tilde{f}^{-1}(b, g) = (b, \rho(b)^{-1}g),$$

which is also continuous. Thus $\tilde{f}$ is an isomorphism of principal G-bundles. This proves the proposition for trivial bundles.

Since every principal G-bundle is locally trivial, the general case follows immediately: for any $b \in B$, choose a trivializing neighborhood U containing b, and apply the above argument to the restrictions $P|_U$ and $Q|_U$. Patching the local inverses together yields a global inverse, so $\tilde{f}$ is an isomorphism. □

Definition 1.4 (Pullback bundle)**.** *Let $\pi : E \to Y$ be a principal G-bundle and $f : X \to Y$ a continuous map. The **pullback bundle** f^*E is defined as:*

$$f^*E = \{(x, e) \in X \times E \mid f(x) = \pi(e)\}.$$

The projection $\tilde{\pi} : f^*E \to X$ is given by:

$$\tilde{\pi}(x, e) = x.$$

The right G-action on f^*E is defined by:

$$(x, e) \cdot g = (x, e \cdot g).$$

Theorem 1.5 (Covering Homotopy Theorem)**.** *Let $\pi : E \to B$ and $\tilde{\pi} : \tilde{E} \to \tilde{B}$ be two fiber bundles with the same fiber type F, and let $\tilde{f} : E \to \tilde{E}$ be a bundle map over $f : B \to \tilde{B}$, i.e., $\tilde{\pi} \circ \tilde{f} = f \circ \pi$.*

Suppose $H : B \times I \to \tilde{B}$ is a homotopy such that $H_0 = f$. Then there exists a homotopy $\tilde{H} : E \times I \to \tilde{E}$ covering H, i.e., the following diagram commutes:

$$\begin{array}{ccc} E \times I & \xrightarrow{\tilde{H}} & \tilde{E} \\ {\scriptstyle \pi \times \mathrm{id}}\downarrow & & \downarrow{\scriptstyle \tilde{\pi}} \\ B \times I & \xrightarrow[H]{} & \tilde{B} \end{array}$$

Moreover, $\tilde{H}_0 = \tilde{f}$.

Theorem 1.6. *Let $\pi : E \to B$ be a principal G-bundle, and suppose $f_0, f_1 : X \to B$ are homotopic maps. Then the pullback bundles are isomorphic:*

$$f_0^*E \cong f_1^*E$$

as principal G-bundles over X.

Proof. Let $H : X \times I \to B$ be a homotopy between f_0 and f_1, so $H_0 = f_0$ and $H_1 = f_1$. Consider the pullback bundle $p_1 : f_0^*E \to X$ and the original bundle $\pi : E \to B$. These have the same fiber type (the group G), and there is a natural bundle map $\tilde{f_0} : f_0^*E \to E$ over f_0 given by projection onto the second factor:

$$\tilde{f_0}(x, e) = e.$$

By the Covering Homotopy Theorem, there exists a homotopy $\tilde{H} : f_0^*E \times I \to E$ such that $\pi \circ \tilde{H} = H \circ (p_1 \times \mathrm{id})$, i.e., the following diagram commutes:

$$\begin{array}{ccc} f_0^*E \times I & \xrightarrow{\tilde{H}} & E \\ {\scriptstyle p_1 \times \mathrm{id}}\downarrow & & \downarrow{\scriptstyle \pi} \\ X \times I & \xrightarrow[H]{} & B \end{array}$$

By the universal property of the pullback bundle, $\tilde{H}$ induces a bundle morphism $\Phi : f_0^*E \times I \to H^*E$ over the identity map on $X \times I$:

$$\begin{array}{ccc} f_0^*E \times I & \xrightarrow{\Phi} & H^*E \\ \downarrow & & \downarrow \\ X \times I & \xrightarrow[\mathrm{id}]{} & X \times I \end{array}$$

Since Φ is a morphism of principal G-bundles over the same base with identity base map, it is an isomorphism by Proposition 1.3.

Restricting to $t = 0$ and $t = 1$, we obtain:

$$\Phi_0 : f_0^*E \xrightarrow{\cong} H_0^*E = f_0^*E, \quad \Phi_1 : f_0^*E \xrightarrow{\cong} H_1^*E = f_1^*E.$$

Thus $f_0^*E \cong f_1^*E$ as principal G-bundles over X. □

Definition 1.7. *A principal G-bundle $\pi : P \to X$ is called **universal** if the total space P is weakly contractible (i.e., the identity map id_P is homotopic to a constant map).*

Notation: *We write*

$$\pi : EG \to BG$$

*for a universal principal G-bundle. The base space BG is called a **classifying space** for the topological group G, and EG is called the **universal total space**.*

Theorem 1.8 (Classification of principal G-bundles)**.** *Let $\pi : EG \to BG$ be a universal principal G-bundle, and let X be a CW-complex. Then the map*

$$\Phi : [X, BG] \longrightarrow \mathcal{P}\mathrm{rin}_G(X)$$

defined by

$$\Phi([f]) = [f^*(EG)]$$

is a bijection.

Notation:

- $[X, BG]$ *denotes the set of homotopy classes of continuous maps* $f : X \to BG$,
- $\mathcal{P}\mathrm{rin}_G(X)$ *denotes the set of isomorphism classes of principal* G*-bundles over* X,
- $f^*(EG)$ *denotes the pullback bundle of* EG *along* f.

Notation:

- $\mathrm{Hom}_G(P, X)$ denotes the set of G-equivariant maps $P \to X$.
- $\Gamma(B, P \times_G X)$ denotes the set of sections of the associated bundle $q : P \times_G X \to B$.
- The associated bundle is $P \times_G X = (P \times X)/G$, where G acts diagonally by $(p, x) \cdot g = (p \cdot g, x \cdot g)$.

Let $\pi : P \to B$ be a principal G-bundle, X a right G-space, and $f : P \to X$ a G-equivariant map. Then the map $P \to P \times X$ given by $p \mapsto (p, f(p))$ is also G-equivariant, and passing to G-orbits yields a map $s = s_f : B \to P \times_G X$ that is in fact a section of the fiber bundle map $q : P \times_G X \to B$.

Write $\mathrm{Hom}_G(P, X)$ for the set of G-equivariant maps $P \to X$, and $\Gamma_{P,X}$ for the set of sections of q.

Proposition 1.9. *Let* $\pi : P \to B$ *be a principal* G*-bundle, and let* X *be a right* G*-space. Then there is a natural bijection*

$$\phi : \mathrm{Hom}_G(P, X) \longrightarrow \Gamma(B, P \times_G X)$$

given by $f \mapsto s_f$*, where* $s_f(b) = [p, f(p)]$ *for any* $p \in \pi^{-1}(b)$.

Proof. We prove the bijection first for trivial bundles, then extend to the general case via local triviality.

Case 1: Trivial bundle. Suppose $P = B \times G$ is the trivial principal G-bundle, with projection $\pi(b, g) = b$ and right G-action $(b, g) \cdot h = (b, gh)$.
Step 1: Equivariant maps. A map $f : B \times G \to X$ is G-equivariant if

$$f(b, gh) = f(b, g) \cdot h \quad \text{for all } b \in B,\ g, h \in G.$$

In particular, taking $g = e$, we have $f(b, h) = f(b, e) \cdot h$. Thus f is completely determined by its restriction to $B \times \{e\}$. Define $\tilde{f} : B \to X$ by $\tilde{f}(b) = f(b, e)$. Then:

$$f(b, g) = \tilde{f}(b) \cdot g.$$

Conversely, any continuous map $\tilde{f} : B \to X$ defines a G-equivariant map $f(b, g) = \tilde{f}(b) \cdot g$. Therefore,

$$\mathrm{Hom}_G(B \times G, X) \cong \mathrm{Hom}(B, X).$$

Step 2: Sections of the associated bundle. The associated bundle is:

$$(B \times G) \times_G X = (B \times G \times X)/G,$$

where G acts by $(b, g, x) \cdot h = (b, gh, x \cdot h)$. The map

$$\psi : B \times X \longrightarrow (B \times G) \times_G X, \quad (b, x) \longmapsto [b, e, x]$$

is a homeomorphism, with inverse given by $[b, g, x] \mapsto (b, x \cdot g^{-1})$. Under this identification, the projection $q : (B \times G) \times_G X \to B$ becomes the standard projection $\pi_1 : B \times X \to B$, $(b, x) \mapsto b$.

A section $s : B \to B \times X$ of π_1 has the form $s(b) = (b, \tilde{s}(b))$ for a unique continuous map $\tilde{s} : B \to X$. Thus,

$$\Gamma(B, (B \times G) \times_G X) \cong \Gamma(B, B \times X) \cong \mathrm{Hom}(B, X).$$

Step 3: The bijection ϕ. Under the above identifications, an equivariant map $f(b, g) = \tilde{f}(b) \cdot g$ corresponds to $\tilde{f} \in \mathrm{Hom}(B, X)$. The section s_f is given by:

$$s_f(b) = [(b, e), f(b, e)] = [(b, e), \tilde{f}(b)] \in (B \times G) \times_G X.$$

Via the homeomorphism ψ^{-1}, this becomes $(b, \tilde{f}(b)) \in B \times X$, which corresponds to $\tilde{f}$ as a section. Thus ϕ is a bijection for trivial bundles.

Case 2: General bundle. Let $\{U_i\}_{i \in I}$ be an open cover of B such that the restrictions $P_i = P|_{U_i}$ are trivial bundles. There is an evident commutative diagram of sets:

$$\begin{array}{ccccc}
\mathrm{Hom}_G(P, X) & \longrightarrow & \prod_i \mathrm{Hom}_G(P_i, X) & \rightrightarrows & \prod_{i,j} \mathrm{Hom}_G(P_i \cap P_j, X) \\
\downarrow{\scriptstyle \phi} & & \downarrow{\scriptstyle \prod \phi_i} & & \downarrow{\scriptstyle \prod \phi_{ij}} \\
\Gamma(P \times_G X \to B) & \longrightarrow & \prod_i \Gamma(P_i \times_G X \to U_i) & \rightrightarrows & \prod_{i,j} \Gamma((P_i \cap P_j) \times_G X \to U_i \cap U_j)
\end{array}$$

Explanation of the diagram:

- For sets, the **equalizer** of two parallel maps $f, g : A \rightrightarrows B$ is defined as:

$$\mathrm{Eq}(f, g) = \{a \in A \mid f(a) = g(a)\}.$$

- Saying that the rows are **exact** means that the first arrow in each row is an isomorphism onto the equalizer of the two subsequent parallel arrows. This is the set-theoretic analogue of exactness in abelian categories, expressing the sheaf condition (gluing data).

The vertical maps are:

- $\phi : \mathrm{Hom}_G(P, X) \to \Gamma(P \times_G X \to B)$, the global bijection we want to establish.
- $\prod \phi_i : \prod \mathrm{Hom}_G(P_i, X) \to \prod \Gamma(P_i \times_G X \to U_i)$, which is a bijection since each P_i is trivial (by Case 1).
- $\prod \phi_{ij} : \prod \mathrm{Hom}_G(P_i \cap P_j, X) \to \prod \Gamma((P_i \cap P_j) \times_G X \to U_i \cap U_j)$, which is injective (again by Case 1, applied to the overlaps).

A diagram chase reminiscent of the Five Lemma shows that for any commutative diagram of this form with:

- exact rows (first arrow is equalizer of the double arrows),
- the middle vertical map $\prod \phi_i$ bijective,
- the right vertical map $\prod \phi_{ij}$ injective,

the left vertical map ϕ must be bijective.

Proof of the diagram chase:

- **Injectivity of ϕ:** Suppose $\phi(f_1) = \phi(f_2)$. Then their images under the left horizontal maps agree in $\prod \Gamma(P_i \times_G X \to U_i)$. Since $\prod \phi_i$ is bijective, the corresponding elements in $\prod \mathrm{Hom}_G(P_i, X)$ are equal. By exactness of the top row, this implies $f_1 = f_2$.
- **Surjectivity of ϕ:** Let $s \in \Gamma(P \times_G X \to B)$. Its image under the bottom left horizontal map is an element of $\prod \Gamma(P_i \times_G X \to U_i)$. Since $\prod \phi_i$ is bijective, this comes from a unique element in $\prod \mathrm{Hom}_G(P_i, X)$. The two parallel arrows map this element to elements in $\prod \mathrm{Hom}_G(P_i \cap P_j, X)$ whose images under $\prod \phi_{ij}$ agree (since s is globally defined). By injectivity of $\prod \phi_{ij}$, the two parallel images are equal, so the element lies in the equalizer. By exactness of the top row, it comes from a global $f \in \mathrm{Hom}_G(P, X)$. Then $\phi(f) = s$ by commutativity.

Thus ϕ is bijective for arbitrary principal G-bundles. □

Corollary 1.10. *Suppose B is a CW-complex and $i : A \hookrightarrow B$ is the inclusion of a subcomplex. Let $\pi : P \to B$ be a principal G-bundle and X a right G-space. If either:*

1. *$i : A \hookrightarrow B$ is a weak homotopy equivalence, or*
2. *X is weakly contractible,*

then any G-equivariant map $f : P|_A \to X$ extends to a G-equivariant map $\tilde{f} : P \to X$.

Proof. By Proposition 1.9, there is a natural bijection between G-equivariant maps $P \to X$ and sections of the associated fiber bundle $q : P \times_G X \to B$. Under this correspondence, the given G-equivariant map $f : P|_A \to X$ corresponds to a section $s_f : A \to P \times_G X$ of q over A.

The projection $q : P \times_G X \to B$ is a fiber bundle with fiber X. Since B is a CW-complex, any fiber bundle over B is a Serre fibration. Thus q is a Serre fibration.

Now consider the extension problem for sections:

$$\begin{array}{ccc} A & \xrightarrow{s_f} & P \times_G X \\ {\scriptstyle i}\big\downarrow & \overset{\exists\, \tilde{s}}{\nearrow} & \big\downarrow{\scriptstyle q} \\ B & \xrightarrow[\mathrm{id}]{} & B \end{array}$$

We have a section defined on the subspace A, and we seek an extension to a global section $\tilde{s} : B \to P \times_G X$.

By standard obstruction theory for Serre fibrations over CW-complexes:

- If the inclusion $i : A \hookrightarrow B$ is a weak homotopy equivalence, then the obstruction to extending the section vanishes, and an extension $\tilde{s}$ exists.
- If the fiber X is weakly contractible (i.e., $\pi_n(X, x_0) = 0$ for all $n \geq 0$), then all obstruction classes lie in zero groups, so any section over A extends to a global section regardless of the inclusion i.

In either case, a global section $\tilde{s} : B \to P \times_G X$ extending s_f exists. Applying the inverse of the bijection in Proposition 6.1, this section corresponds to a G-equivariant map $\tilde{f} : P \to X$ that extends the original map f. □

Proof of Theorem 1.8. Suppose EG is weakly contractible. We prove that the map

$$\Phi : [X, BG] \longrightarrow \mathcal{B}un_G(X), \quad [f] \longmapsto [f^*(EG)]$$

is bijective.

Surjectivity: Let $Q \to X$ be an arbitrary principal G-bundle over X. We must show there exists a map $f : X \to BG$ such that $f^*(EG) \cong Q$.

Apply Corollary 1.10 to the CW-pair $(X, \emptyset)$ with the G-space $X' = EG$. Since EG is weakly contractible, condition (2) of Corollary 1.10 is satisfied. Therefore, the empty G-equivariant map from $Q|_\emptyset$ to EG extends to a G-equivariant map $\tilde{f} : Q \to EG$.

Passing to G-orbits yields a continuous map $f : X = Q/G \to EG/G = BG$ on the base spaces. This gives a morphism of principal G-bundles over X:

$$\begin{array}{ccc} Q & \xrightarrow{\tilde{f}} & EG \\ \downarrow & & \downarrow \\ X & \xrightarrow[f]{} & BG \end{array}$$

By the universal property of the pullback, this morphism factors through $f^*(EG)$, yielding a bundle map $Q \to f^*(EG)$ over X. Since any morphism of principal G-bundles over the same base with identity base map is an isomorphism (Proposition 1.6), we conclude $Q \cong f^*(EG)$. Thus $\Phi([f]) = [Q]$, proving surjectivity.

Injectivity: Suppose $f_0, f_1 : X \to BG$ are maps such that $\Phi([f_0]) = \Phi([f_1])$, i.e., there exists an isomorphism of principal G-bundles over X:

$$\psi : f_0^*(EG) \xrightarrow{\cong} f_1^*(EG).$$

We must show that f_0 and f_1 are homotopic.

Set $Q = f_0^*(EG)$. Then f_0 and f_1 are both classifying maps for the same bundle Q, meaning there exist G-equivariant maps $\tilde{f_0} : Q \to EG$ and $\tilde{f_1} : Q \to EG$ covering f_0 and f_1 respectively. Specifically:

- $\tilde{f_0} : Q \to EG$ is the natural projection onto the second factor,

- $\tilde{f_1} : Q \to EG$ is the composition $Q \xrightarrow{\psi} f_1^*(EG) \xrightarrow{\pi_2} EG$.

Consider the product bundle $R = Q \times I \to X \times I$, which is a principal G-bundle over $X \times I$. The restrictions to the ends are:

$$R|_{X\times\{0\}} \cong Q, \quad R|_{X\times\{1\}} \cong Q.$$

The maps $\tilde{f_0}$ and $\tilde{f_1}$ together define a G-equivariant map on the subspace $X \times \{0, 1\}$:

$$R|_{X\times\{0,1\}} \longrightarrow EG.$$

Apply Corollary 1.10 to the CW-pair $(X \times I, X \times \{0, 1\})$. Since EG is weakly contractible, this G-equivariant map extends to a G-equivariant map on the whole product:

$$\tilde{H} : R = Q \times I \longrightarrow EG.$$

Passing to G-orbits yields a continuous map:

$$H : X \times I = (Q \times I)/G \longrightarrow EG/G = BG.$$

By construction, $H_0 = f_0$ and $H_1 = f_1$, so H is a homotopy between f_0 and f_1. Thus $[f_0] = [f_1]$ in $[X, BG]$, proving injectivity.

Therefore, Φ is a bijection. □

Proposition 1.11. *Suppose a universal principal G-bundle $\pi : AG \to BG$ exists. Then:*

(a) BG can be taken to be a CW-complex;

(b) a CW-classifying space BG is unique up to canonical homotopy equivalence;

(c) EG is unique up to G-homotopy equivalence.

Proof. **(a)** For any space BG, we can choose a CW-complex B' and a weak homotopy equivalence $g : B' \to BG$. Consider the pullback bundle $g^*EG \to B'$. We claim g^*P is weakly contractible.

The projection map $g^*P \to P$ induces a map of Serre fibrations:

$$\begin{array}{ccc} g^*EG & \longrightarrow & EG \\ \downarrow & & \downarrow \\ B' & \xrightarrow[g]{} & BG \end{array}$$

This induces a LES of homotopy groups:

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & \pi_n(g^*EG) & \longrightarrow & \pi_n(B') & \longrightarrow & \pi_{n-1}(G) & \longrightarrow & \cdots \\ & & \downarrow & & \downarrow{\scriptstyle g_*} & & \| & & \\ \cdots & \longrightarrow & \pi_n(EG) & \longrightarrow & \pi_n(BG) & \longrightarrow & \pi_{n-1}(G) & \longrightarrow & \cdots \end{array}$$

Since EG is weakly contractible, $\pi_n(EG) = 0$ for all $n \geq 0$. Since $g : B' \to BG$ is a weak equivalence, $g_* : \pi_n(B') \to \pi_n(BG)$ is an isomorphism for all n. By the Five Lemma applied to the exact sequences, the map $\pi_n(g^*EG) \to \pi_n(EG) = 0$ is an isomorphism, so $\pi_n(g^*EG) = 0$ for all n. Thus g^*EG is weakly contractible, and $g^*EG \to B'$ is a universal principal G-bundle with base a CW-complex.

(b) By Theorem 1.8 , the classifying space B represents the functor

$$\mathcal{P}\mathrm{rin}_G(()-) : \mathbf{hCW}^{\mathrm{op}} \longrightarrow \mathbf{Sets}$$

from the homotopy category of CW-complexes to sets. That is, for any CW-complex X, there is a natural bijection:

$$[X, B] \cong \mathcal{P}\mathrm{rin}_G(()X).$$

If B_1 and B_2 are two CW-classifying spaces, then they both represent the same functor $\mathcal{B}un_G(-)$. By Yoneda's Lemma, any two objects representing the same functor in a category are canonically isomorphic. In the homotopy category of CW-complexes, isomorphism means homotopy equivalence. Thus B_1 and B_2 are canonically homotopy equivalent.

□

For this theory to be of any use, we need to know that classifying spaces exist. The following theorem is due to Milnor.

Theorem 1.12 (Milnor)**.** *Let G be any topological group. Then there exists a classifying space BG for G, i.e., a universal principal G-bundle $EG \to BG$.*

Since $\pi : EG \to BG$ is a principal G-bundle, it is a Serre fibration as BG can be taken to be CW compelex hence paracompact. The fiber over any point $b \in BG$ is homeomorphic to the group G:

$$\pi^{-1}(b) \cong G.$$

The long exact sequence of homotopy groups associated to this fibration is:

$$\cdots \longrightarrow \pi_n(G) \longrightarrow \pi_n(EG) \longrightarrow \pi_n(BG) \longrightarrow \pi_{n-1}(G) \longrightarrow \pi_{n-1}(EG) \longrightarrow \cdots$$

Since EG is weakly contractible (universal), we have:

$$\pi_n(EG) = 0 \quad \text{for all } n \geq 0.$$

Substituting this into the long exact sequence, we obtain for each $n \geq 1$:

$$0 \longrightarrow \pi_n(BG) \xrightarrow{\cong} \pi_{n-1}(G) \longrightarrow 0$$

Therefore, the boundary map gives an isomorphism:

$$\pi_n(BG) \cong \pi_{n-1}(G) \quad \text{for all } n \geq 1.$$

For $n = 1$, the exact sequence at the end gives:

$$\cdots \longrightarrow \pi_1(EG) \longrightarrow \pi_1(BG) \longrightarrow \pi_0(G) \longrightarrow \pi_0(EG) \longrightarrow \pi_0(BG) \longrightarrow 0.$$

Since $\pi_1(EG) = 0$ and $\pi_0(EG) = 0$ (as EG is path-connected and contractible), we have:

$$\pi_1(BG) \cong \pi_0(G), \quad \pi_0(BG) = 0.$$

Thus BG is path-connected, and its fundamental group is the group of path components of G.

Special Case: G is a discrete group. If G is a discrete topological group, then:

$$\pi_0(G) = G, \quad \pi_n(G) = 0 \quad \text{for all } n \geq 1.$$

From the isomorphism $\pi_n(BG) \cong \pi_{n-1}(G)$, we deduce:

$$\pi_1(BG) \cong G, \quad \pi_n(BG) = 0 \quad \text{for all } n \geq 2.$$

Therefore, the classifying space BG is an Eilenberg–MacLane space $K(G, 1)$.

Conclusion

We have established the classical classification theorem for principal G-bundles:

> For a discrete group G and a CW-complex X, there is a natural bijective correspondence:
>
> $$\mathcal{P}\mathrm{rin}_G(X) \xrightarrow{\cong} [X, K(G, 1)]$$
>
> given by pulling back the universal principal G-bundle $EG \to BG = K(G, 1)$.

Thus, isomorphism classes of principal G-bundles over X are completely classified by homotopy classes of maps from X to the Eilenberg–MacLane space $K(G, 1)$.

This correspondence provides a powerful bridge between the geometric theory of fiber bundles and the homotopy-theoretic study of classifying spaces, forming a cornerstone of algebraic topology.